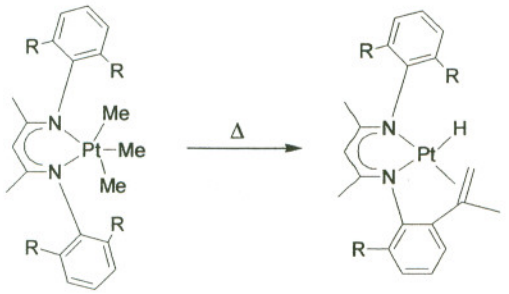
Inorganic Cume
February 27, 2004

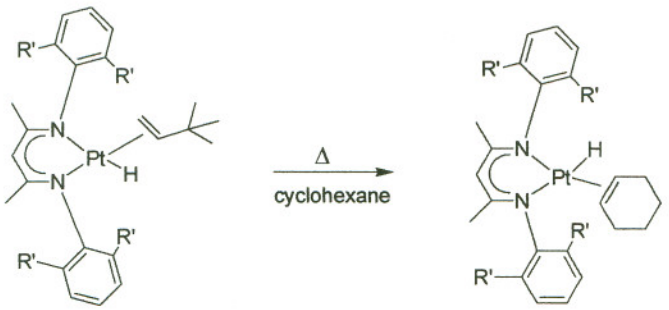
1) In a recent *JACS* report, Goldberg and co-workers reported the below reaction. Propose a mechanism for the reaction shown in equation 1.



(1)

R = i-Pr

2) In the same paper they also reported the below reaction in which they also changed the nature of their aryl ligands. Propose a mechanism for the reaction shown in equation 2.



(2)

1

R' = CH_3

3) Attempts to catalytically dehydrogenate cyclohexane to cyclohexene were unsuccessful. It was found that a substitution of cyclohexene from compound **1** with the hydrogen acceptor t-butyl ethylene did not occur, even under forcing conditions such as adding a large excess of the t-butyl ethylene. Assuming that olefin substitution on Pt(II) is normally an associative process, why do you think substitution did not occur?

4) Draw the catalytic cycle that you think the authors were trying to achieve.